Multimodal Large Language Model

Multimodal = Model which can handle more than one data type.

Key Terms in the Language Model Ecosystem
What is an MLLM (Multimodal Large Language Model) or what is Multimodality? - **Clear Definition**
Models examples: Open Source, Close Source, Leaderboards
Why Do LLMs Need Multi Modalities?
What is Multimodal LLM Fine-Tuning? When and Why Should You Use Multimodal Fine-Tuning?
Architecture Overview
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Normal LLM Fine-Tuning vs Multimodal Fine-Tuning - **Clear Comparison**
Practical Using Unsloth

Key Terms in the Language Model Ecosystem

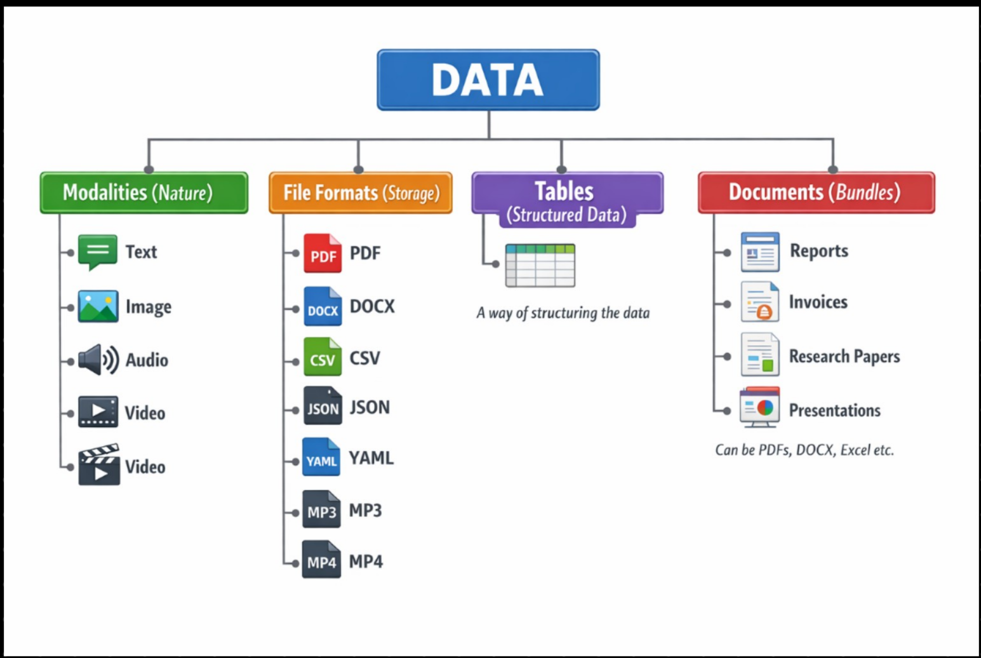
LLM (Large Language Model)
A Transformer-based model trained on massive datasets to understand and generate language.
Examples: GPTs, Claudes, Geminis, LLaMAs etc...

MLLM (Multimodal Large Language Model)
An LLM extended with additional capability (vision/audio/video).
Examples: GPT-4o, Gemini 2.5, Qwen3-VL, LLaVA etc...
Vision-Language Model (VLM)
Audio-Language Model or Speech Model

SLM (Small Language Model)
A compact language model or Multimodal (typically <10B parameters) optimized for efficiency, speed, and specialization.
Examples: Phi-3 Mini, Gemma 2B, Gemma Vision, Tiny Llama, Qwen2-VL-2B, LLaVA Small

Closed-Source
- API-based
- No weight access
- Limited fine-tuning access

Open-Source
- Weight access(download possible)
- Fine-tuning possible
- Local deployment



1. What is an MLLM (Multimodal Large Language Model) or what is Multimodality? (Clear Definition)

A multimodal LLM is an AI model that can understand, process and generate multiple types of data (modalities).

Examples of modalities:

- 📝 Text
- 🖼️ Images
- 🎧 Audio
- 🎥 Video

Multimodal LLM

Modality Group	Transformation Type	Description / Use Case	Closed-Source Models	Open-Source Models
Text Modality	Text → Text	Classic LLM case.	Open AI GPT models, Anthropic Claude, Google Gemini etc...	Llama, Mistrals etc...
Image Modality	Image → Text	Provide an image and extract the content.	Open AI GPT-4o & 5, Google DeepMind Gemini 1.5 / 2.0, Anthropic Claude 3 Vision	Salesforce BLIP-2, Meta LLaVA, Alibaba Group Qwen-VL, Google Gemma Vision, Hugging Face IDEFICS
Image Modality	Text → Image	Generate an image from a text prompt.	OpenAI DALL·E, Midjourney, Gemini-2.5-flash-image, Gemini-3 image preview (Nano banana)	Stability AI: Stable Diffusion, Black Forest Labs: FLUX.x
Image Modality	Image → Image	Modify, enhance, or transform an input image. (Background removal, Style transfer, Super resolution, Face swap etc...)	Adobe Firefly (image editing), Open AI DALL·E (image editing), Nano banana (image editing)	Stability AI: Stable Diffusion (img2img / inpainting)
Audio Modality	Audio → Text	Convert spoken audio into text (Speech Recognition).	OpenAI Whisper API, Google Speech-to-Text	OpenAI Whisper (open model), Mozilla DeepSpeech
Audio Modality	Text → Audio	Convert text into synthesized speech (Text-to-Speech).	OpenAI OpenAI TTS, Google Text-to-Speech	Coqui AI TTS, ESPnet TTS
Audio Modality	Audio → Audio	Transform or modify input audio.	ElevenLabs Voice AI, Adobe Enhance Speech	Meta Voice box
Video Modality	Video = Images + Time + Audio	Video modality combines images + time + audio.	—	—
Video Modality	Text → Video	Generate video from text prompt.	Runway Gen-2, Pika Labs, Open AI SORA	Stability AI Stable Video Diffusion, ModelScope Text-to-Video
Video Modality	Video → Text	Generate description or reasoning from video input. Examples: Video captioning, Scene understanding, Action recognition	Open AI GPT-4o, Gemini Vision Model	Alibaba Group Qwen-VL (video variants), Qwen3-Omni, OpenGVLab InternVideo
Video Modality	Video → Video	Modify or transform existing video. Examples: Style transfer, Video enhancement, Face swap	Runway Video editing tools, Adobe Firefly Video	Stability AI Stable Video Diffusion (img2vid / vid2vid), OpenMMLab / MMEditing

https://huggingface.co/spaces/open-llm-leaderboard/open_llm_leaderboard#/
https://huggingface.co/spaces/lmarena-ai/lmarena-leaderboard
https://huggingface.co/papers/trending

Why Do LLMs Need Multiple Modalities?
Since the Real World Is Multimodal
Humans don't understand the world through text alone.
We use:

- Language (text, speech)
- Audio (tone, emotion)
- Vision (images, video)

and build the Context (environment)

Models need multiple modalities because the real world is not text-only — intelligence requires seeing, hearing, and understanding context .

Example:
A doctor looks at:

- X-ray image
- Patient report
- Lab values

All together — not just text.

If AI only understands text, it sees only a small part of reality.

What is Multimodal LLM Fine-Tuning?
Adapting a MM Large Language Model to perform a specific task(Domain Specific Task) using multimodal data. (image + text, audio + text, video + text)

When Should You do Multimodal Fine-Tuning?
Multimodal understanding
OCR reasoning
Visual conversation
Document AI

Multimodal LLM fine-tuning is the process of adapting a MM large language model that processes both visual and textual inputs by updating its projection layers and/or language model weights using image-text instruction datasets to improve domain-specific cross-modal reasoning.

ViT: https://arxiv.org/pdf/2010.11929
CLIP: https://arxiv.org/pdf/2103.00020
Qwen-VL: https://arxiv.org/pdf/2308.12966
Qwen-VL-2.5: https://arxiv.org/pdf/2502.13923
BLIP-2: https://arxiv.org/abs/2301.12597
LLAVA: https://arxiv.org/abs/2304.08485
openAI Dalle: https://arxiv.org/pdf/2103.00020
Google Imagen: https://arxiv.org/pdf/2205.11487

Architecture overview

Image modality = Image to text, Text to Image

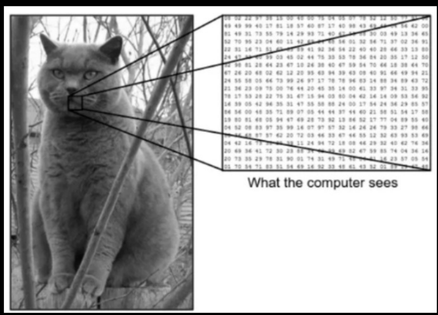
What is an Image mathematically?

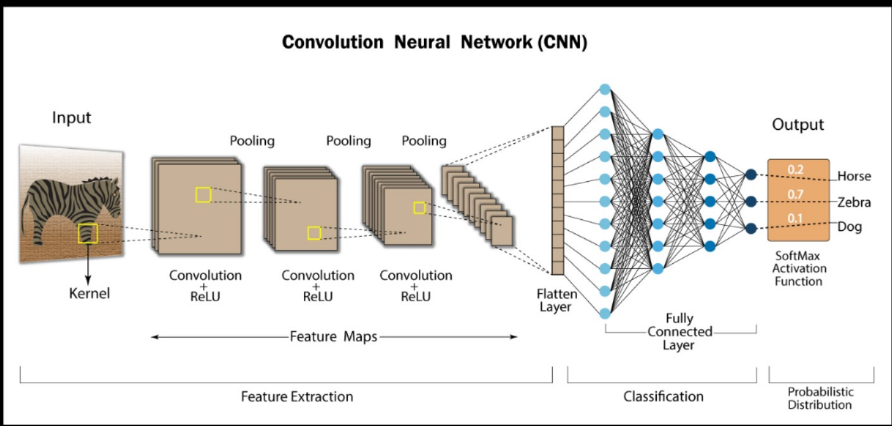
CNN Basics

Google Vision Transformer (ViT)

CLIP, LLaVA, Qwen(image to text)

Dall-E, Imagen(Nano Banana) (text to image)





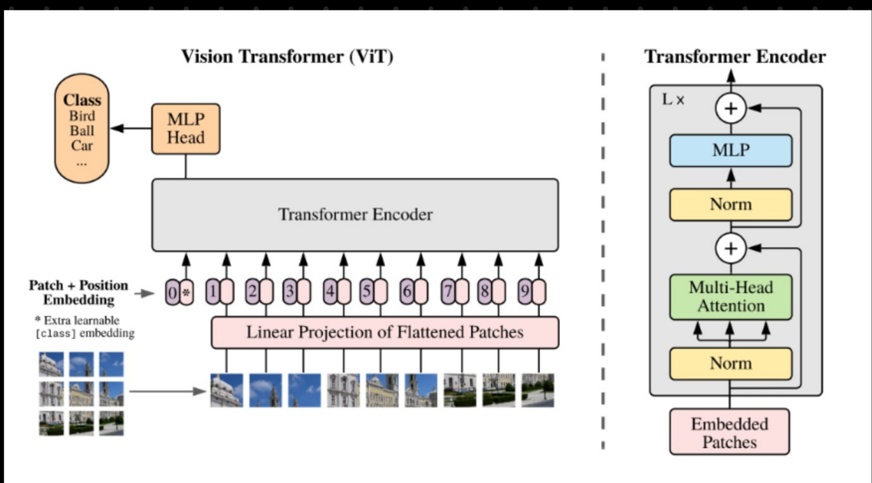
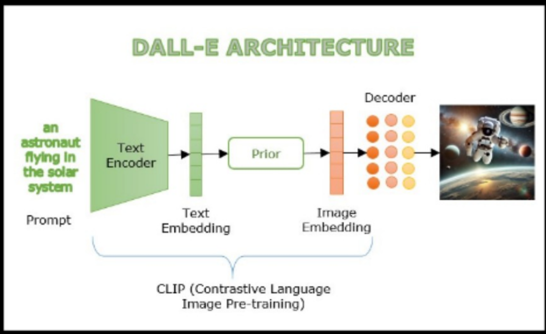


Image
↓
Vision Encoder (CLIP / ViT)
↓
Image features (vector)
↓
Projection Layer
↓
Converted embeddings
↓
LLM
↓
Text output

Core idea (ViT): The paper proves you can do image recognition with a pure Transformer (no CNN) by splitting an image into fixed-size patches (e.g., 16×16), flattening each patch into a vector, linearly embedding them like “tokens,” adding positional embeddings, and feeding the sequence into a standard Transformer encoder



User Prompt
↓
CLIP Text Encoder
↓
Text Embedding
↓
Prior Model (Text → Image embedding)
↓
Image Embedding
↓
Diffusion Decoder (Noise → Image)
↓
Final Generated Image

What Exactly Gets Fine-Tuned?
There are multiple strategies in multimodal fine-tuning:

Option 1: Train Everything (Very Expensive)
Train:Vision encoder, Projection, LLM
Used by large labs with massive compute
Requires very large dataset + multi-node GPU clusters
Not practical for individual creators.

Option 2: Train Only Projection Layer (Cheapest)
Freeze: Vision encoder, LLM
Train: Projection layer (vision → LLM embedding mapper)
Used in early LLaVA (v1) style setups
Purpose: Teach LLM how to interpret visual embeddings
Limitation: Weak reasoning improvement because LLM remains fully frozen.

Option 3: Train Projection + LLM (Common)
Freeze: Vision encoder
Train: Projection layer, LLM (LoRA or full fine-tuning)
This is the most widely used approach today
Improves multimodal reasoning significantly
Most open-source multimodal models follow this.

Option 4: LoRA-based Multimodal Fine-Tuning (Practical)
✔ Vision Layers → LoRA applied
✔ LLM Layers(Attention, MLP) → LoRA applied

Means:
You are NOT fully training them.
You are training LoRA adapters inside:
This:
Works on ~24GB GPU (depending on model size)
Memory efficient

Dataset format: Dataset format is different from text-only training.

- Example 1: Simple Image + Instruction

```
{
"image":"example.png",
"text" or "caption": "I am sunny"
}
```

- Example 2: VQA Format

```
{
  "image": "chart.png",
  "question": "What is the highest value?",
  "answer": "The highest value is 72."
}
```

- Example 3: LLaVA Format

```
{
  "image": "image_001.jpg",
  "conversations": [
    {"role": "user", "content": "What is in this image?"},
    {"role": "assistant", "content": "A dog sitting on a sofa."}
  ]
}
```

```
{
  "id": "0001",
  "image": "image1.jpg",
  "conversations": [
    {"from": "human", "value": "<image>\nDescribe the scene."},
    {"from": "gpt", "value": "A busy street with cars and pedestrians."}
  ]
}
```

Note: <image> token tells model where visual embedding goes.

Aspect	**Normal LLM Fine-Tuning**	**Multimodal LLM Fine-Tuning**
Input	Text only	Image/Audio/Video + Text
Architecture	LLM only	Encoder + Projection + LLM
Compute	Medium	Medium to High
GPU Need	Moderate VRAM	Higher VRAM (often)
Complexity	Medium	High
Preprocessing	Tokenization	Media processing + Tokenization
Stability	More stable	More sensitive